

CULLMAN 3 ENE, AL, USA

WMO: 720010

Lat: 34.1954N

Lon: 86.7963W

Elev: 244

StdP: 98.43

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63867

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.0	(c) -5.6	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 7.7	(k) 8.1	(l) 6.8	(m) 7.9	(n) 2.2	(o) 10	(p) 0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.6	(c) 33.9	(d) N/A	(e) 32.7	(f) N/A	(g) 31.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 2.6	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(a) 6.3	(b) 5.4	(c) 4.7	(e) -11.7	(f) 35.3	(g) 2.6	(h) 1.8	(i) -13.5	(j) 36.5	(k) -15.0	(l) 37.6	(m) -16.5	(n) 38.6	(o) -18.3	(p) 39.8		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	4.6	7.1	11.3	15.4	19.9	24.2	25.3	25.3	22.3	16.0	9.9	7.4
	DBStd	8.44	6.00	5.79	5.20	4.13	3.51	1.99	1.85	2.05	3.13	4.59	4.66	5.34
	HDD10.0	539	188	114	48	7	1	0	0	0	0	6	57	118
	HDD18.3	1793	426	314	221	104	27	0	0	0	6	101	254	340
	CDD10.0	2649	21	35	89	169	306	427	476	473	370	193	54	37
	CDD18.3	862	0	1	3	16	75	177	217	214	126	29	2	1
	CDH23.3	7700	0	3	37	162	686	1665	1932	1882	1053	268	10	1
	CDH26.7	2932	0	0	3	13	182	683	798	807	384	61	1	0
(14) Wind	WSAvg	2.1	2.4	2.6	2.5	2.4	1.9	1.8	1.7	1.6	1.6	1.9	2.1	2.3
(15) Precipitation	PrecAvg	1558	147	142	165	131	136	117	125	94	129	92	124	149
	PrecMax	2083	275	353	461	338	310	318	261	222	361	259	327	363
	PrecMin	1011	32	12	36	11	42	14	23	9	2	3	44	31
	PrecStd	244	64	72	80	68	64	66	60	51	74	62	54	80
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.7	23.5	26.9	28.3	31.6	34.5	35.1	36.3	33.7	30.4	24.6	22.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	18.4	20.7	24.1	26.7	30.1	33.0	33.3	34.0	32.1	28.2	21.9	20.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	16.5	19.1	22.2	25.4	28.8	31.8	32.1	32.4	30.7	26.4	20.2	18.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	14.5	16.9	20.2	23.7	27.4	30.6	30.9	31.0	29.1	24.4	18.6	16.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	11.1	12.0	13.0	12.0	11.9	10.6	10.8	11.6	12.7	12.6	10.3
		MCDBR	11.9	12.9	14.0	14.3	13.9	13.8	12.4	13.0	13.4	14.8	14.2	11.3
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.315	0.321	0.361	0.394	0.445	0.475	0.518	0.520	0.426	0.373	0.337	0.317
	taud		2.510	2.492	2.398	2.295	2.196	2.147	2.048	2.030	2.283	2.416	2.459	2.521
	Ebn at Noon		897	930	912	893	847	818	780	771	835	857	860	873
	Edh at Noon		84	94	113	131	146	153	168	169	125	101	87	79
(44) All-Sky Solar Radiation	RadAvg		2.52	3.04	4.20	5.42	5.93	6.21	5.95	5.52	4.80	3.92	2.87	2.15
	RadStd		0.19	0.39	0.39	0.38	0.46	0.49	0.35	0.30	0.42	0.51	0.31	0.21

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	+105

Nomenclature: See separate page